
MA 512: Linear Algebra

Course Code	MA 512
Course Name	Linear Algebra
L-T-P-C	3-1-0-4
Prerequisites	Basic knowledge on matrix and determinants
Intended for	M.Sc./M.S./Ph.D.
Distribution	Core for M.Sc. in applied Mathematics and Elective for other discipline
Approval	10th Senate (Jun 2016)

Course Contents

- **Matrices, vectors, and systems of linear equations** [3 Lectures]
Introduction to Matrix and Determinant.
- **Vector spaces, basis, dimension** [10 Lectures] Vector spaces, Subspaces, Subspaces connected with matrices, Linear span, Linear independence, Bases and dimension, Basis and dimension of range and null space.
- **Linear transformations, change of basis** [6 Lectures]
Linear transformations and matrices, Coordinate change, Change of basis and similarity.
- **Diagonalisation** [7 Lectures]
Eigenvalues and eigenvectors, Diagonalisation of a square matrix, Inner products, orthogonality, orthogonal diagonalisation, Applications of diagonalisation.
- **Direct sums and projections** [7 Lectures]
The direct sum of two subspaces, Orthogonal complements, Projections, Characterising projections and orthogonal projections, Minimising the distance to a subspace.
- **Complex matrices, vector spaces** [9 Lectures]
Complex vector spaces, Complex inner product spaces, The adjoint. Hermitian and unitary matrices, Unitary diagonalisation. Normal matrices, Spectral decomposition.

Text Books

1. G.Strang, **Linear Algebra and its Applications**, 4th Edition, Thomson, 2006.
2. K. Hoffman and R. Kunze, **Linear Algebra**, Prentice Hall, 2008.
3. H.Anton, **Elementary Linear Algebra with Applications**, 9th Edition, John Wiley, 2004.

References

1. Loehr, Nicholas, **Advanced Linear Algebra**, Taylor & Francis Inc.
2. Iuliana Iatan, **Advanced Lectures on Linear Algebra with Applications**, LAP Lambert Academic Publishing.
3. Sohail A. Dianat, Eli Saber, **Advanced Linear Algebra for Engineers with MATLAB**, Taylor Francis Inc.